# Topic 1: Introduction to HCI (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia



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01 OVERVIEW OF HCI

02 EVOLUTION OF HCI

03 HISTORY & FUTURE OF HCI



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# OVERVIEW OF HCI



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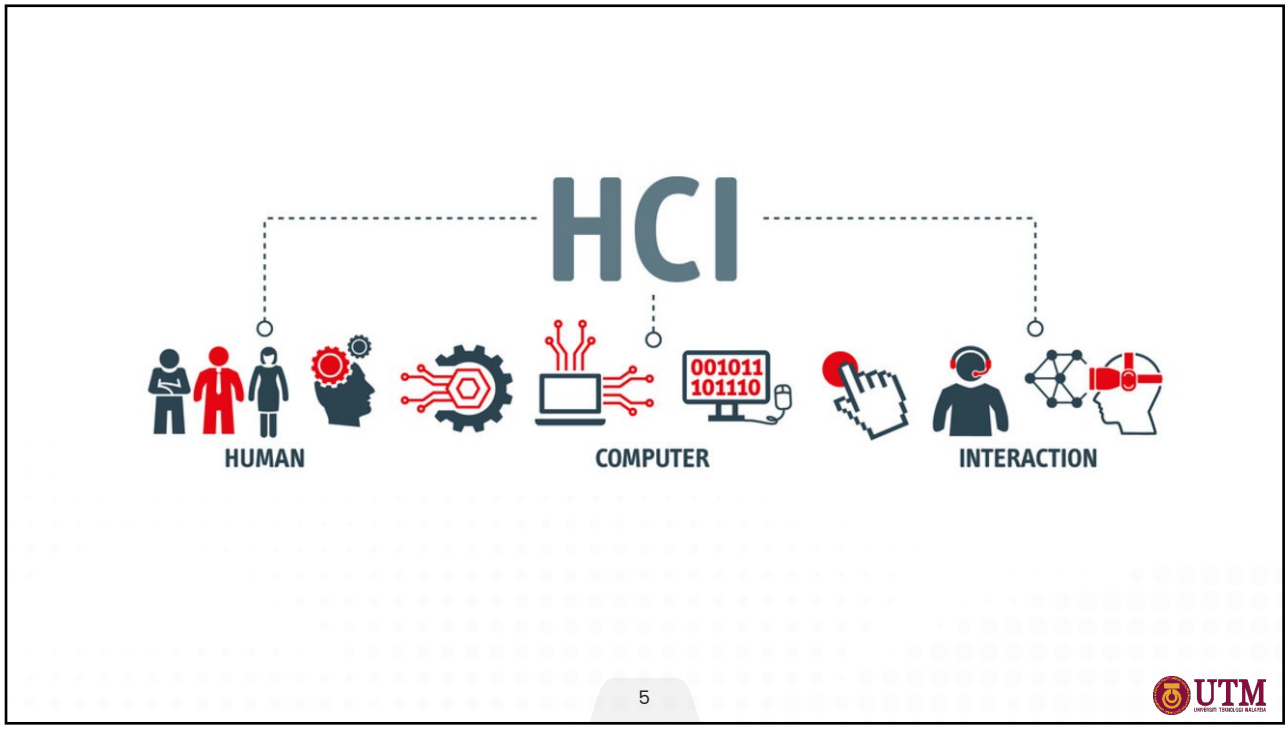
## What is HCI?



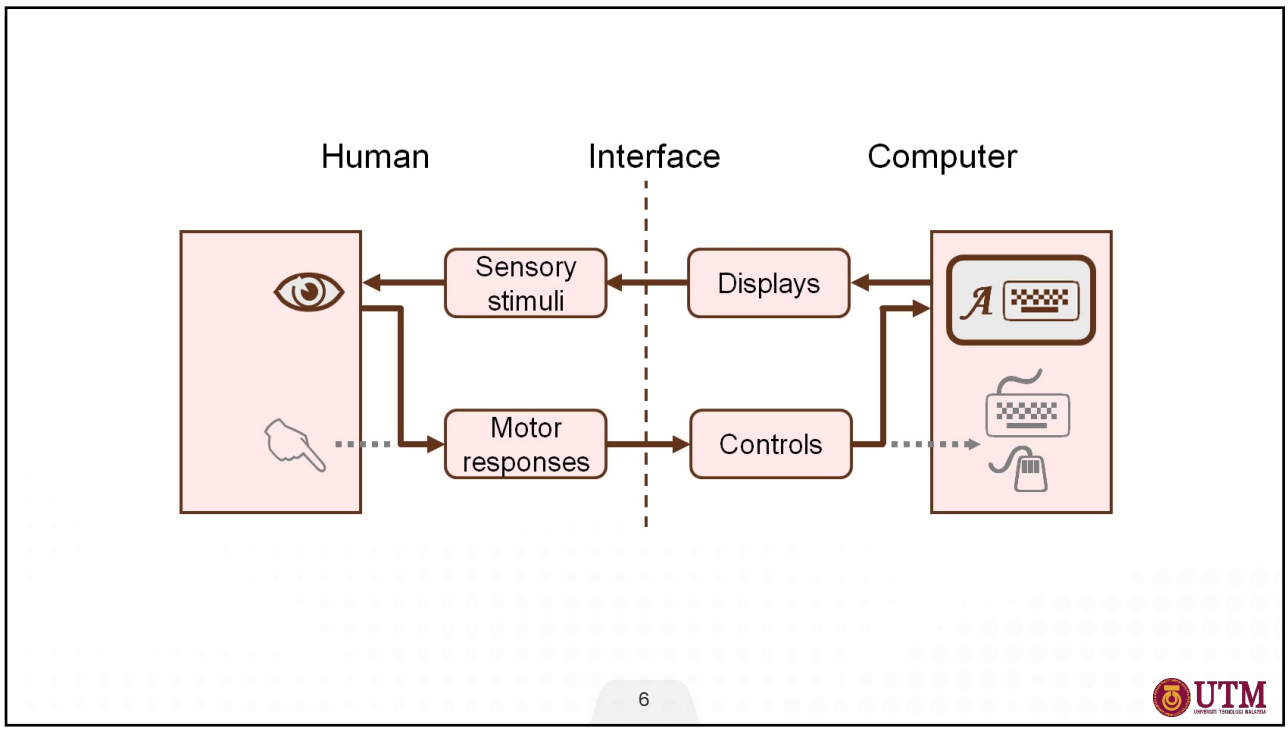

https://youtu.be/yNzLBI0wsGU?si=0g16ZUp8SZPYvHRX

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HCI
001011
101110
HUMAN
COMPUTER
INTERACTION
5
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5


Human
Interface
Computer
Sensory stimuli
Displays
Motor responses
Controls
6
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# What is HCI?

| HUMAN | COMPUTER | INTERACTION |
|---|---|---|
| • Single user<br>• Groups<br>• I/O channels<br>• Memory<br>• Reasoning<br>• Problem solving<br>• Error<br>• Psychology | • Desktop<br>• Embedded system<br>• Data entry devices<br>• Output devices<br>• Memory<br>• Processing | • Direct/Indirect communication<br>• Models<br>• Frameworks<br>• Styles<br>• Ergonomics |



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# Why study HCI?



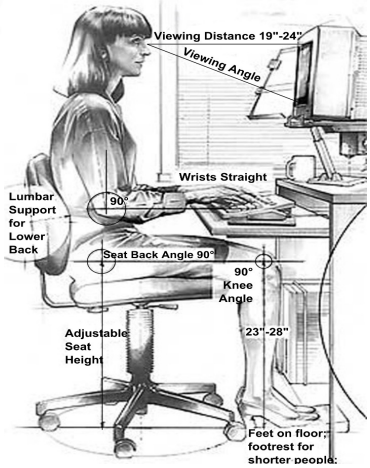

- To employ people more productively and effectively
  - people costs now far outweigh hardware and software costs
- People now expect "easy to use" systems
  - generally they are not tolerant of poorly designed systems
  - if a product is hard to use, they will seek other products



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# Why study HCI?

HUMAN FACTORS VIEW

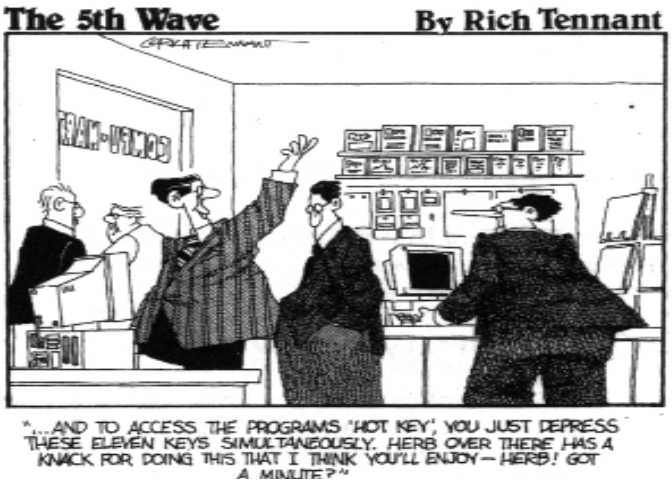



- Human have limitations
- Errors are costly in terms of
  - loss of time & money
  - loss of lives in critical systems
  - loss of morale



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# How do you feel?






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# Why HCI is Important?

- Fail to design well, it can affect
  - Effectiveness
  - Productivity
  - Morale
  - Safety

- Example: a car with poor HCI



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# Activity



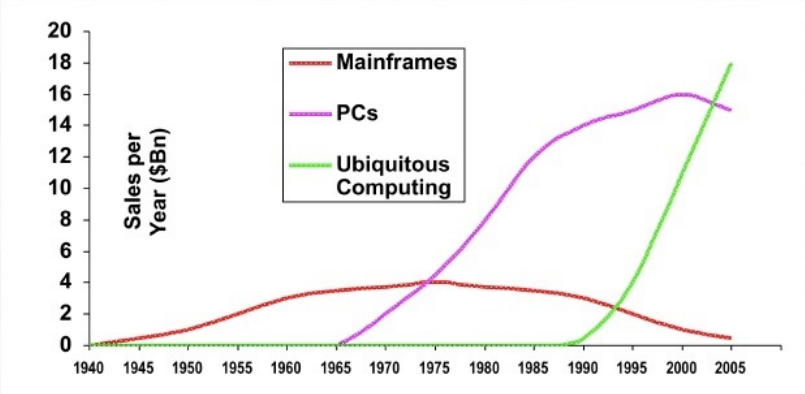
- Take 5 minutes for everyone to write down one common device with substantial HCI design choices and discuss with the neighbor the pros and cons.

- How does it affect you or other users?



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# Weiser's Waves of Computing

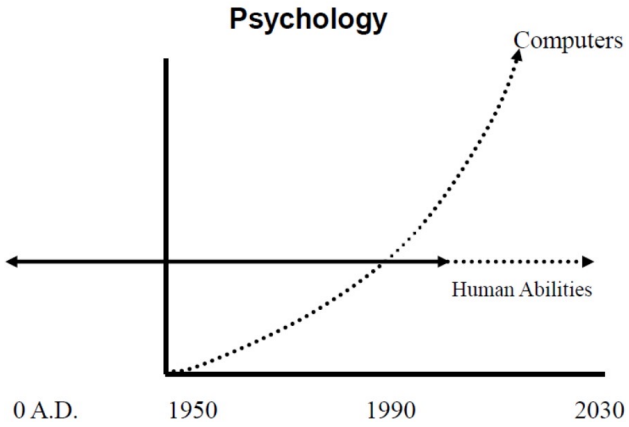



The Three Waves of Computing, Mark Weiser, 1990

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# Human vs. Computer




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# Definition of HCI (1)

"Human-computer interaction is a discipline concerned with the **design, evaluation** and **implementation** of **interactive computing system** for **human use** and with the study of major phenomena surrounding them."

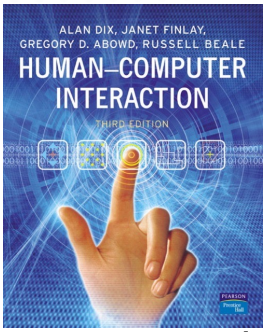
[ACM SIGCHI Curricula Human-Computer Interaction]



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# Definition of HCI (2)

"The study of interaction between people and computer-based systems; concern with the physical, psychological and theoretical aspects of this process."



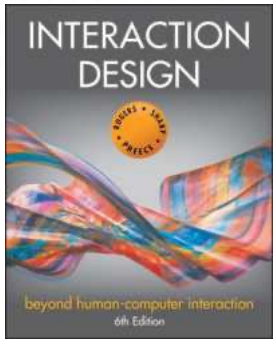

[Alan Dix et al. 2003]



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# Definition of HCI (3)

""Human-computer interaction (HCI) is about designing computer system that support people so that they can carry out their activities productively and safely."




[6th edn Jenny Preece et al. 2023/2024]



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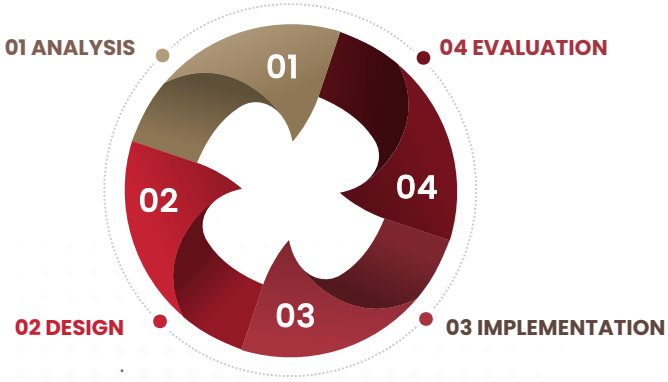
# Definition of HCI (cont.)

- HCI has a **role** in the design and development of all kinds of systems, ranging from those like:
  - air traffic control and nuclear processing where **safety** is extremely important,
  - to office systems, where **productivity** and job satisfaction are paramount,
  - to computer games, which **must excite and engage users**.



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# Components of HCI




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# Goal of HCI

To develop or improve the safety, utility and effectiveness of systems that include computers, often through improving usability

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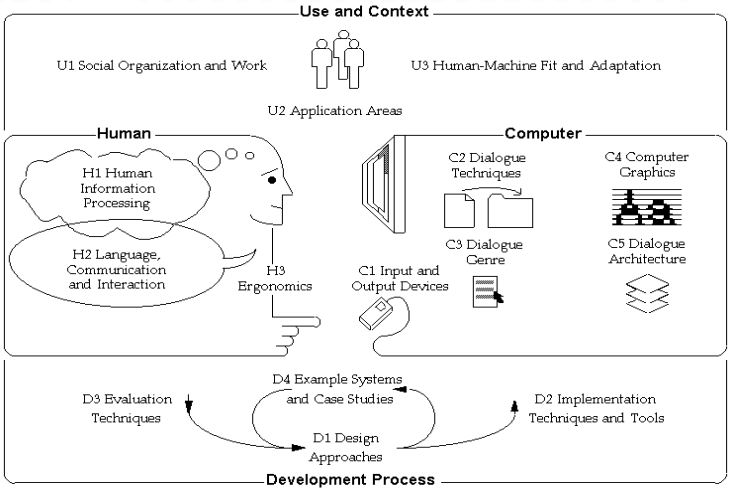
# Field of HCI

- Psychology
- Sociology
- Cognitive science
- Ergonomics / Human factors
- Computer science
- Visualization
- Design
- Interactive art
- Information science
- Information security
- Speech-language pathology
- Personal information management
- Phenomenology
- …





# Meta Models of HCI




The Discipline of HCI [Source: Book "**Human Computer Interaction" (1994,p.16)**]

# Why HCI is important?

"Human-computer interaction is the kind of discipline which is neither the study of humans, nor the study of technology, but rather the bridging between those two. So you always have to have one eye open to the question: what can the technology do? How can you build it? What are the possibilities? And one eye open to the question: what are people doing and how would this fit in? What would they do with it? If you lose sight of either of those, you fail to design well."

(Terry Winograd, Professor of computing, Stanford University)



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# EVOLUTION OF HCI

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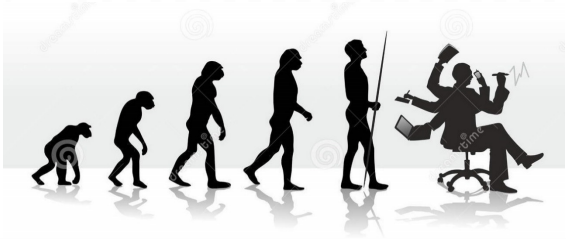
# Mac Evolution






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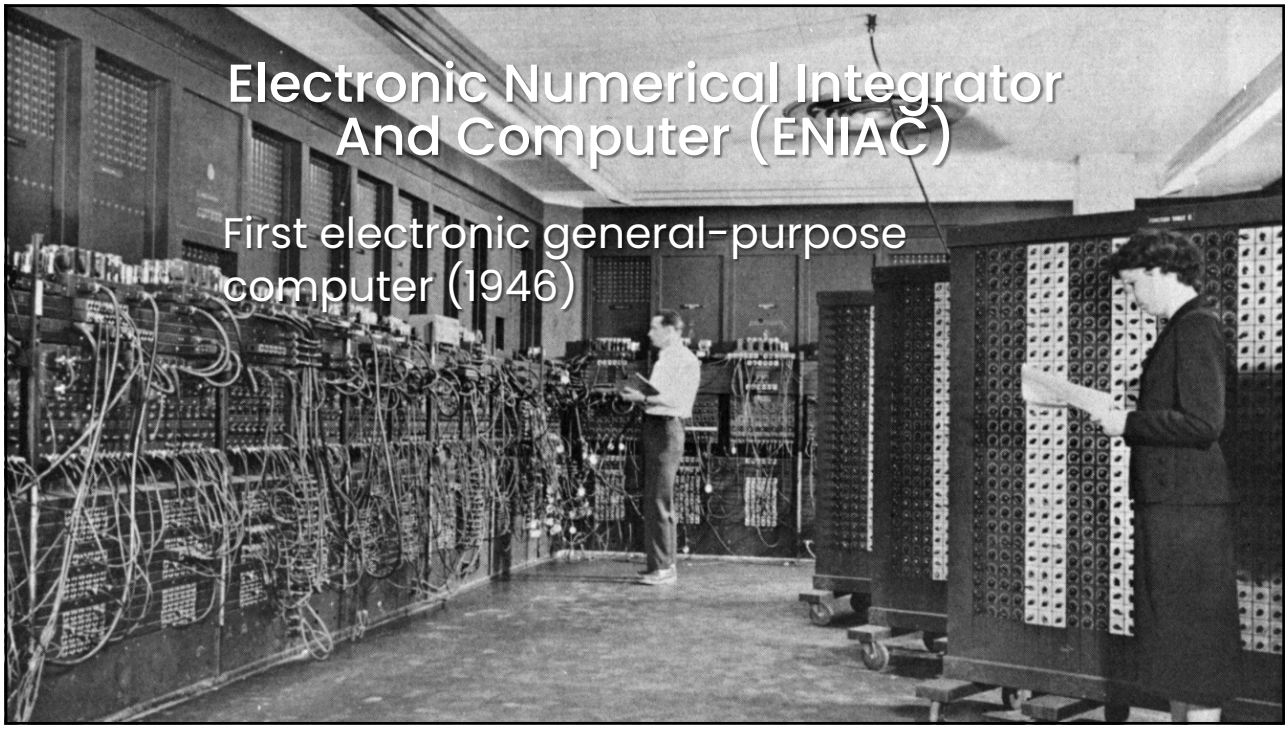
# Human Evolution (?)



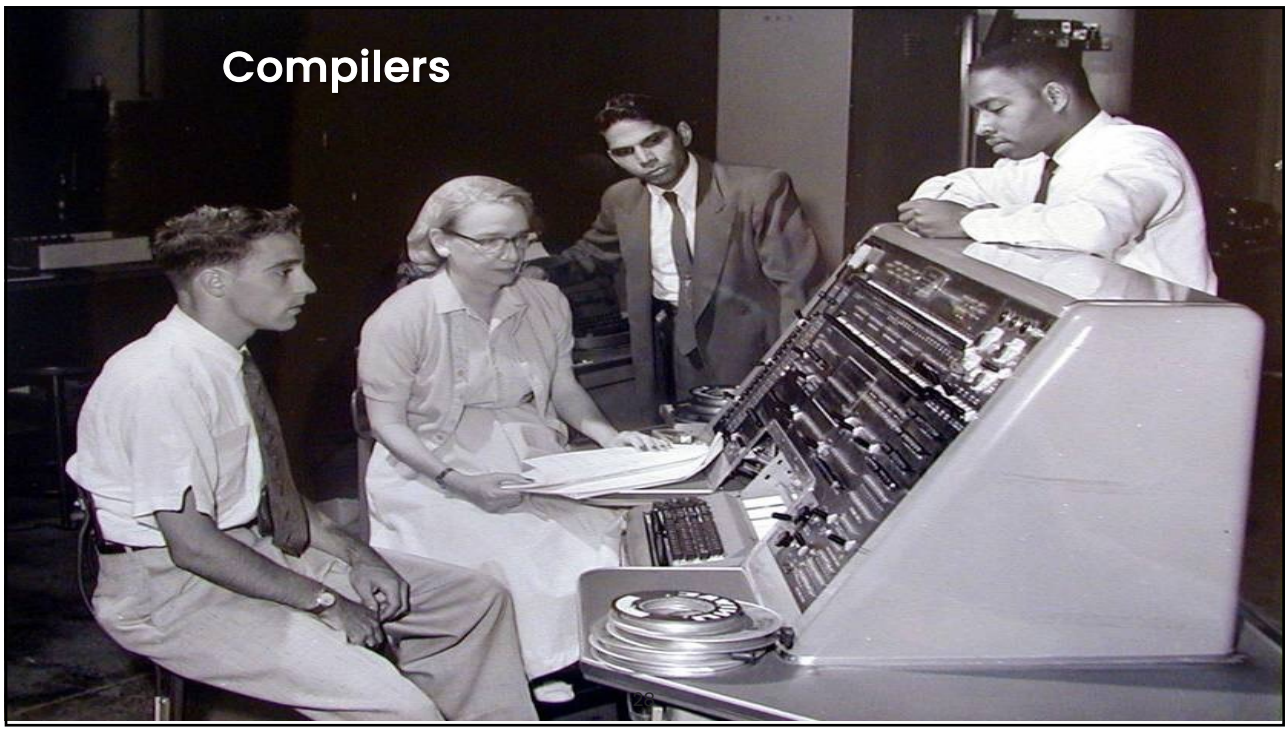





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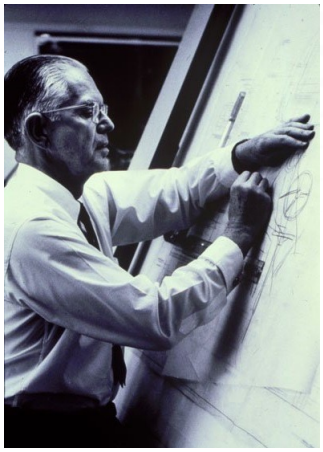

Electronic Numerical Integrator And Computer (ENIAC)
First electronic general-purpose computer (1946)

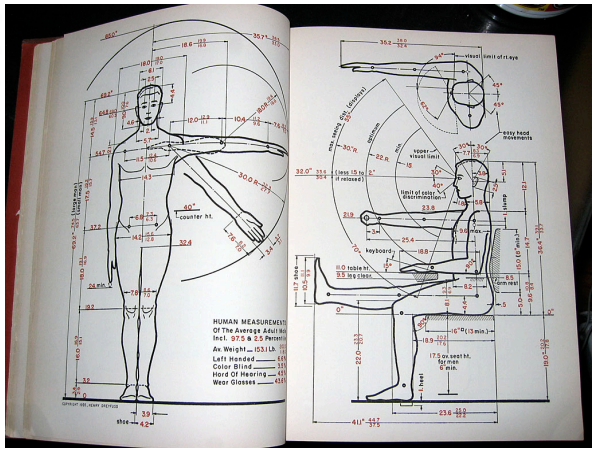
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Compilers

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# Designing for People



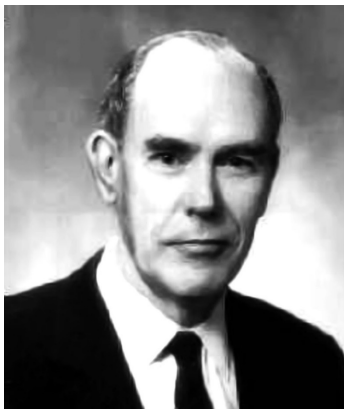


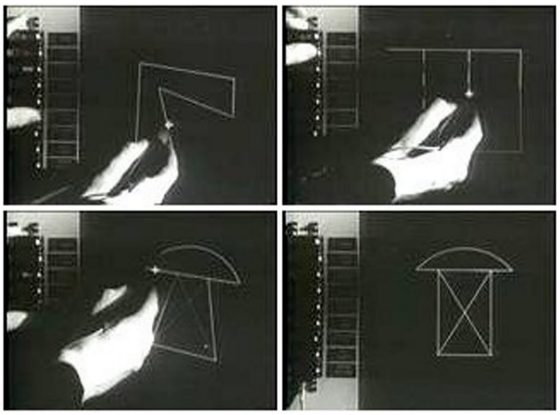
**Henry Dreyfuss** designed specific measurements & proportions of the human body for specific work & environment. [Source: Book "**Designing for People" (1955)**]



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# Graphical User Interface (GUI)



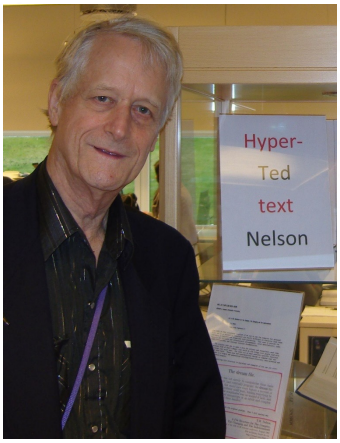


**Ivan Sutherland's Sketchpad (1963)**

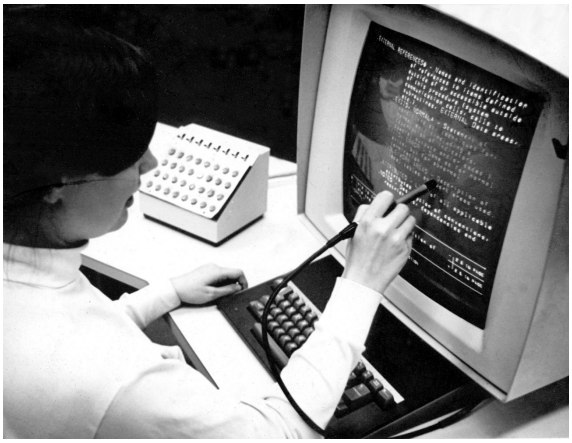


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# Hypertext




**Ted Nelson** introduced the terms "hypertext" & "hypermedia" in 1963



**Hypertext Editing System (HES)** was an early hypertext research project conducted at Brown University in 1967

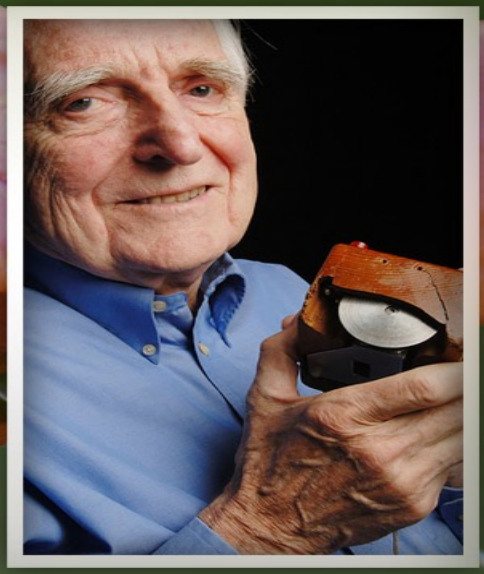
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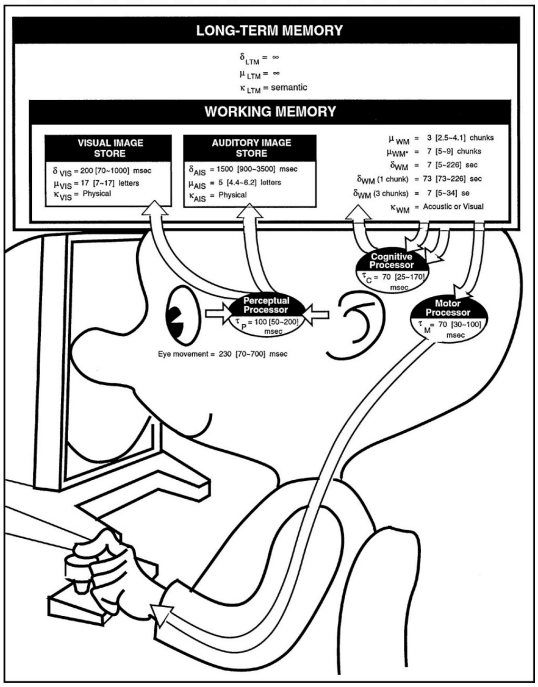
# Mouse

- Douglas Engelbart was an inventor of the computer mouse
- Invented in 1960s



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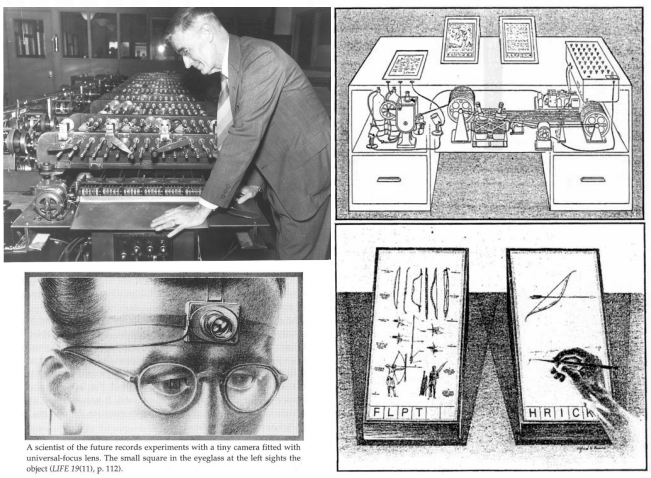

# Model Human Processor

- Introduced by Card, Moran & Newell in 1983
- To calculate how long it takes to perform a certain task using cognitive modeling method





# Memory Extender (MemEx)




A scientist of the future records experiments with a tiny camera fitted with universal-focus lens. The small square in the eyeglass at the left sights the object (*LIFE* 19(11), p. 112).

- A device to store all books, records and communication
- Mechanised to be consulted with exceeding speed and flexibility
- Support for full-text search, text & audio annotations and hyperlinks

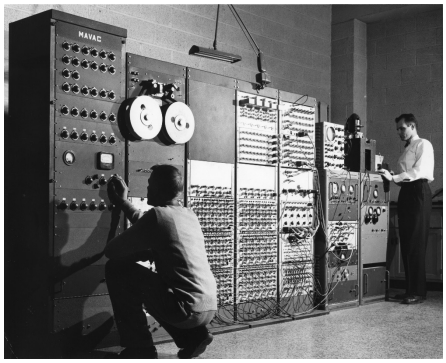
["As We May Think", Vannevar Bush, 1945]

# Man-Computer Symbiosis

- Tightly coupled human brain & machine, speech recognition, time sharing and character recognition

[J.C.R. Licklider, 1960]






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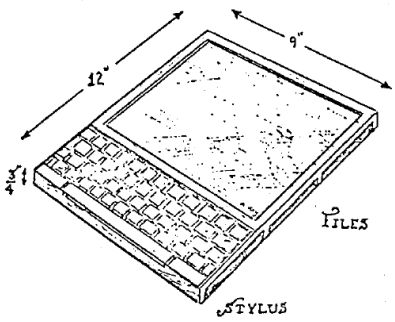
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# DynaBook

The DynaBook

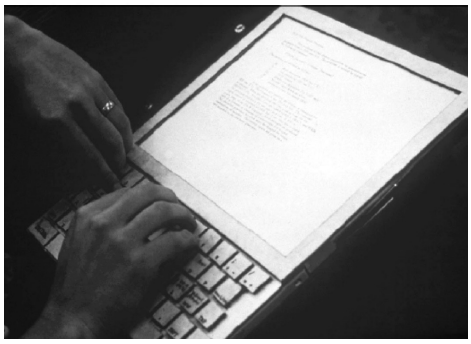
"I wish to God these calculations were executed by steam!"
Charles Babbage (age 19)
ca. 1803

"The Analytical Engine weaves algebraic patterns, just as the Jacquard Loom weaves patterns in silk."
-Ada Augusta
Countess of Lovelace






[Alan Kay, 1968]



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# Xerox Machine



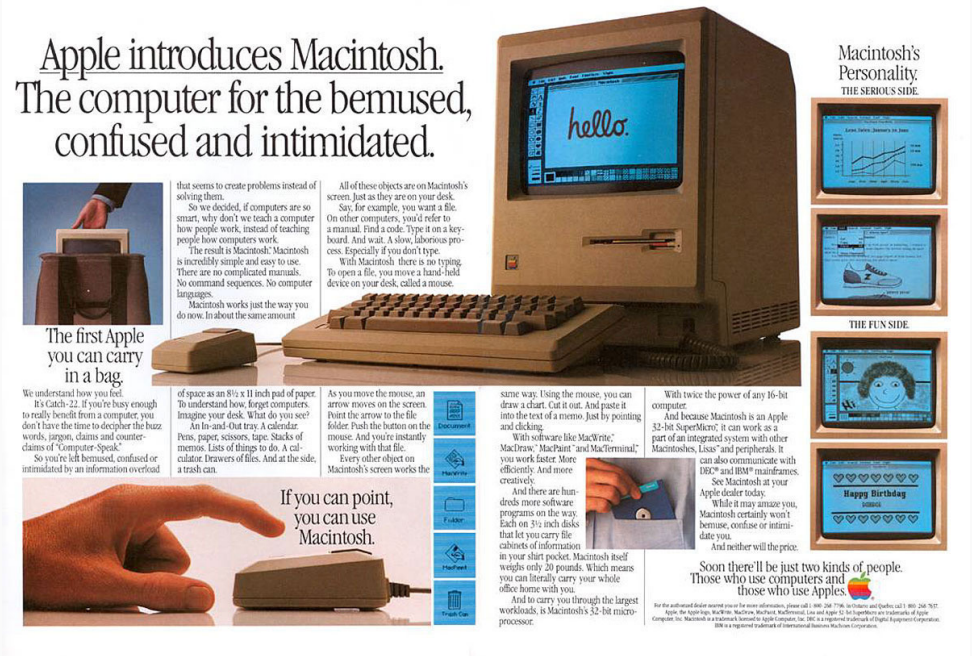
XEROX Alto (1973)



XEROX Star (1981)



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# Challenges to HCI Designers

- Two important challenges to HCI designers:
  - How to keep know what's going on within the development of technology (changes in technology)
  - How to ensure that their designs offer good HCI and take advantage of the potential functionality of the new technology

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# HISTORY & FUTURE OF HCI



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## Where we are today

- Large displays
- Small displays
- Peripheral displays
- Alternative I/O
- Ubiquitous computing
- Virtual environments
- Implants
- Speech recognition
- Multimedia
- Video conferencing
- Artificial intelligence
- Software agents
- Recommender systems
- etc ...



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# The Reactable

- An electronic musical instrument with a tabletop/mobile device
- A multi-touch interface for playing music
- Performers can simultaneously interact by moving and rotating physical objects on its surface
- Use Interfaces Gestures to interact




Source: https://reactable.com/





# MIT's Transform

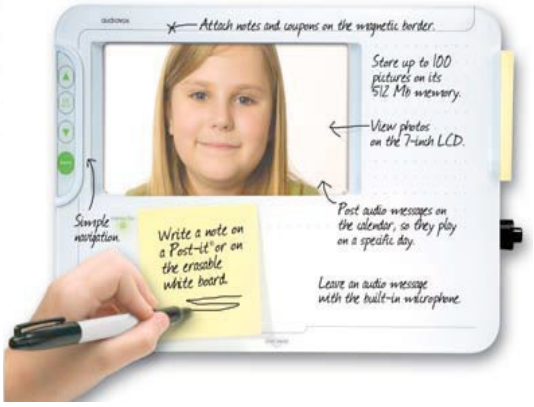
- Dynamic and Adaptive furniture
- It holds and moves objects like fruits, game tokens, office supplies and tablets; creates dividers on demand; and generates interactive sculptures to convey messages and audio.



Source: https://tangible.media.mit.edu/project/transform/

# Audiovox's Digital Message Center

- Digital Message Center is designed to be attached to the refrigerator.
- Letting families scribble digital notes and leave audio and video messages for each other.

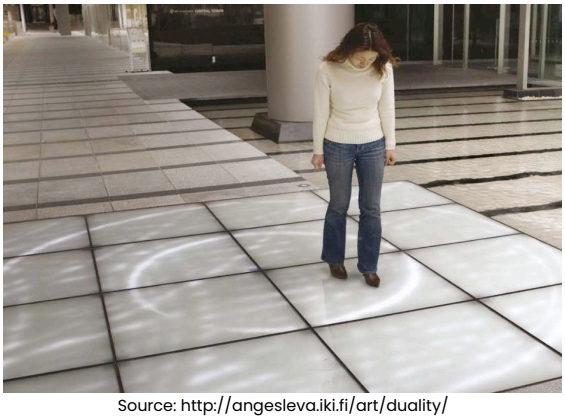





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# Duality

- Duality ART+COM's artistic installation called Duality, located at the exit of a metro station in Tokyo
- Passers-by provoke virtual ripple effects with their footsteps, as if walking across a pond



Source: http://angesleva.iki.fi/art/duality/



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# Mobile Phones Evolution (50 years)



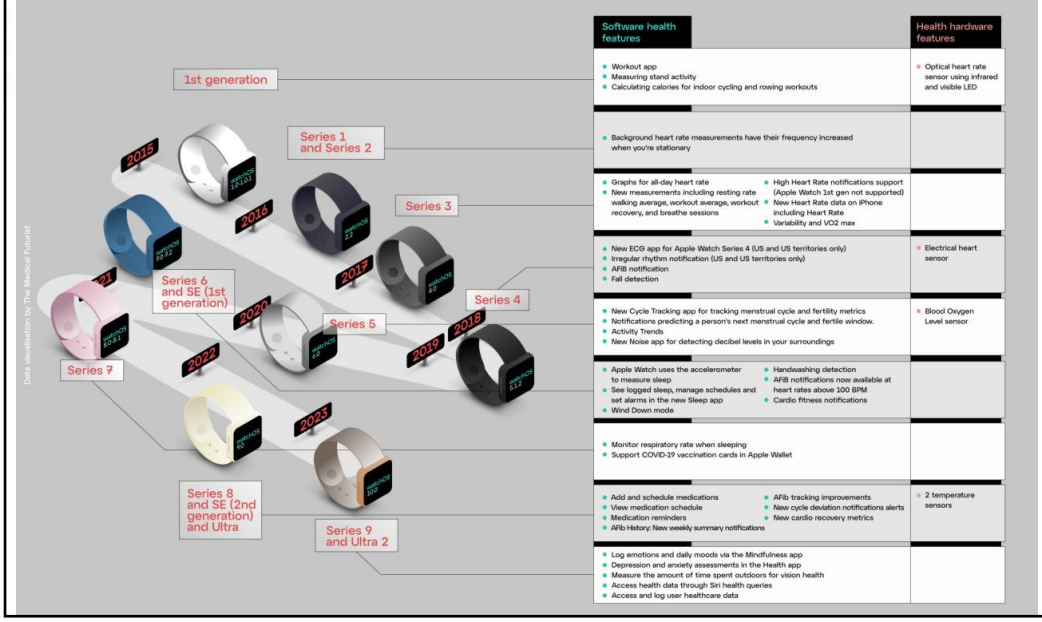

From the first mobile phone 'brick' to the latest Apple iPhone: as the size reduces, the potential expands.

Source: https://www.sellcell.com/blog/first-mobile-phone-call-50th-anniversary-history-of-the-mobile-phone/



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# The Evolution of Apple Watch Health Features




Source: https://cdn.medicalfuturist.com/wp-content/uploads/2023/10/Apple-watch-evolution.jpg

# Smart Living




Source: https://www.tianjian.com.sg/smart-home-solutions





# Film Productions

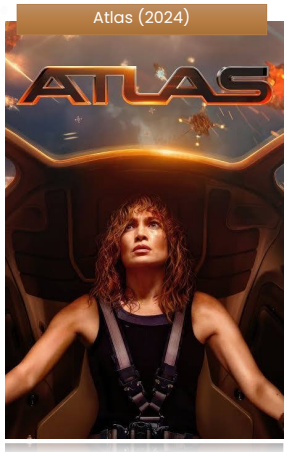
Minority Report (2002)



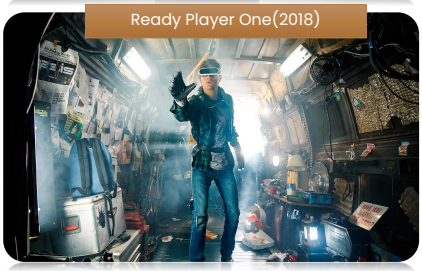
Iron Man (2008)



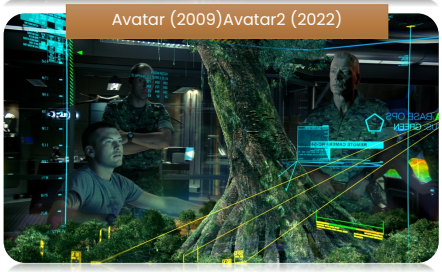
Atlas (2024)




Ready Player One(2018)



Avatar (2009)Avatar2 (2022)






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